

Unsupervised continuous camera network pose estimation through human mesh recovery

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ABSTRACT

Camera resectioning is essential in computer vision and 3D reconstruction to estimate the position of matching pinhole cameras in 3D worlds. While the internal camera parameters are usually known or can be easily computed offline, in camera networks extrinsic parameters need to be computed each time a camera changes position, thus not allowing for smooth and dynamic network reconfiguration. In this work we propose a fully markerless, unsupervised, and automatic tool for the estimation of the extrinsic parameters of a camera network, based on 3D human mesh recovery from RGB videos. We show how it is possible to retrieve, from monocular images and with just a weak prior knowledge of the intrinsic parameters, the real-world position of the cameras in the network, together with the floor plane. Our solution also works with a single RGB camera and allows the user to dynamically add, re-position, or remove cameras from the network.

CCS CONCEPTS

• **Computing methodologies** → **Computer vision; Camera calibration; Epipolar geometry; Artificial intelligence; Image and video acquisition; 3D imaging;**

KEYWORDS

pose estimation, camera calibration, skeleton matching, depth estimation

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1 INTRODUCTION

A lot of effort has been put over the years into automating the process of camera resectioning. The available literature, though, still lacks of a fully unsupervised and markerless approach, which could significantly simplify the deployment of a camera network. Such deficiency in the state of the art might be perhaps due to

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the manifoldness of sensors and lenses on the market, which frustrates any generalization attempt. Another aspect that increases the difficulty of automatic calibration is the dynamic nature of the environments where the camera network is going to be installed. For example, in many scenarios, including video surveillance, Ambient Assisted Living (AAL), environmental monitoring, the reconfiguration of the camera network is generally common, often due to the re-positioning of pieces of furniture, or more in general obstacles that can partially or fully limit the visibility on the scene. In addition, especially when dealing with outdoor scenarios, cameras with PTZ (pan-tilt-zoom) capabilities are often adopted, as they are capable of changing their internal configuration, making it necessary to re-calibrate one or more sensors. Wind or other weather conditions may also be critical, introducing a lot of noise, and making it difficult to accomplish most of the classic vision tasks, such as keypoint extraction, motion detection, and tracking.

Many good solutions to estimate the intrinsic camera parameters to correct lens distortion have been provided in the past, the internal configuration of a camera remains usually fixed, except when zooming or refocusing. On the other hand, extrinsic parameters model the relation between the camera coordinates and the real-world coordinates. For this reason, even a slight movement of the camera can cause a loss in calibration precision. This is problematic because the standard calibration procedures, though not complex, are rather time consuming requiring every time manual intervention.

Many approaches try to simplify the calibration process by reducing the number of required checkerboard images. In autonomous driving, good markerless solutions can be found by using visual odometry [14], SLAM [5], or optical flow [12] for feature tracking; however, they are often not suitable for surveillance cameras, due to the completely different application scenario. Other methods use SIFT/SURF features and feature matching between camera views to estimate the camera parameters [7], but they require additional data from other sensors, such as active range sensors. Recent methods focus on calibrating cameras by using the cues provided by walking humans, both for intrinsic and extrinsic parameters estimation. In particular, these approaches are usually based on:

- Manhattan World Assumption
- Planar trajectories
- Skeleton data from 3D sensors

The ones based on the Manhattan World Assumption [3] usually work well in city environments due to their geometric uniformity, but may fail in all the other scenarios. Human detection and tracking has been used in the past to find vanishing points and estimate the ground plane from multiple camera views [18]. However, they require a prior knowledge of the people or camera height and are not robust to occlusions, noise, or non-standard standing poses.

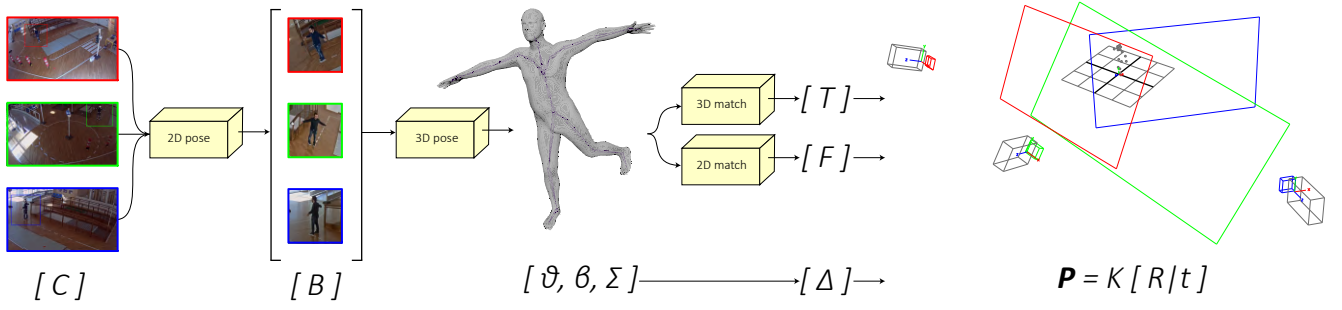


Figure 1: Illustration of the proposed pipeline: from RGB video streams to the estimation of the extrinsic parameters of the camera network.

Recently, RGBD sensors such as the Microsoft Kinect V2k¹ allowed to obtain a better understanding of the scene through depth and 3D skeleton pose estimation. However, the main issues in calibrating a camera network from the skeleton information via Kinect are the small range (<3 m), low precision due to occlusions, and the high infrastructural and processing cost, as multiple computers are required to handle a network of RGBD sensors.

During the last years, researchers in computer vision have spent a lot of efforts in the area of human pose estimation from monocular images. There have been many successful examples, such as [15], [21] and [2]. Many of them have been made possible thanks to the availability of very large datasets, in particular from CMU’s Panoptic Studio [8], leading to popular open source frameworks, such as OpenPose [1]. Among the different kinds of techniques used in the past to extract 3D human pose from monocular videos, we can distinguish between:

- two-stages approaches
- direct approaches

Two-stages approaches, such as [19], first estimate 2D joints and then recover the depth component. On the other hand, direct approaches try to recover the 3D skeleton or mesh in one shot. Among them, Kanazawa et al.’s *End-to-end recovery of human shape and pose* [9] is one of the most known.

Starting from Kanazawa’s work, we extend it to work with multiple views and with a more realistic camera model, that allows for extrinsic parameters estimation. Our results show that, over a small number of frames, the human skeleton alone can provide a sufficient number of keypoints for fully unsupervised camera calibration, to retrieve the real-world 3D position of cameras in a camera network. We show some results with different configurations and discuss on how our method can be further extended for better accuracy.

2 RELATED WORK

Most of the existing related literature is based on the Manhattan World Assumption [3], which assumes that it is easier to find vanishing points from images taken in city environments. While such assumption well suits the city scenario, still it fails in many others especially when considering indoor scenes. Many methods focus

on estimating the position of a new camera inserted as a node into a network of calibrated cameras. Vasconcelos et al. achieve that in [20] by using sets of pairwise correspondences between images. However, that requires having a set of extrinsically calibrated cameras. Hold-Geoffroy et al. in [6] focus on single camera calibration to estimate focal length, pitch and roll using a deep CNN, demonstrating how the advent of deep neural networks can be of help with very good results also in domains traditionally handles using geometrical constraints. With the growing popularity of omnidirectional cameras, Miyata et al. in [13] show how to exploit their large field of view to anchor non-overlapping views. Zhao et al. in [24] explore an augmented reality solution that allows automatic extrinsic calibration by placing AR markers directly on top of cameras. However, both of these methods require the addition of a new camera or a set of markers to the scene.

Vanishing points estimation is perhaps the most popular approach for automatic camera calibration. The authors in [22] propose a creative solution that exploits the geometry of solar panels to estimate orthogonal vanishing points. Tang et al. in ESTHER [18] propose a complete solution for joint intrinsic and extrinsic parameters estimation, based on human trajectories. However, their approach is more focused on intrinsic calibration for distortion correction. Additionally, they require a prior knowledge of the cameras heights and work correctly only in scenes of people walking in normal upright position.

Only other few works are based on human pose cues, and most of them exploit 3D sensors data.

Desai et al. in [4] propose a skeleton-based method for continuous calibration of Kinect V2 sensors. In their work, they also explore some issues related to working with depth sensors, such as low range of vision, skeleton flipping and high computational costs. Among the numerous works in 3D pose estimation, we chose to base our work on Kanazawa et al.’s approach [9], because it provides a richer mesh representation of the human body in terms of both shape and pose. Additionally, it also gives some cues on camera pose for rendering. The limitation we found, is that it is based on a weak perspective camera model, which could be limiting in the scenarios we address. As detailed later, we will improve and extend their method with a more realistic camera model, suitable for calibration.

¹<https://developer.microsoft.com/en-us/windows/kinect>

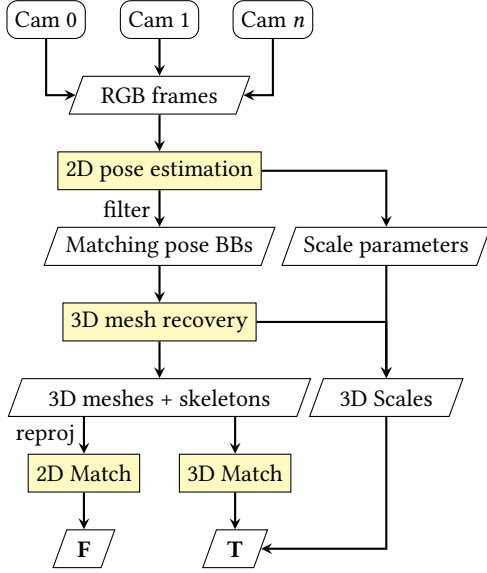


Figure 2: An overview of our camera network calibration pipeline

3 THE PROPOSED MODEL

A simplified overview of our calibration architecture can be seen in figure 2. In short, the proposed method takes as input sequences of RGB frames from an arbitrary number of camera streams $C_0, \dots, C_n$, then applies 2D pose estimation to detect matching skeletons and the corresponding bounding boxes $B_0, \dots, B_n$. We then infer the 3D position of skeletal joints and their real-world scale through human mesh recovery. Finally, we align the skeletons centroids and use a least squares approach to find a set of rigid transformations $T_{\{i \rightarrow 0 | i=1, \dots, n\}}$ from each skeleton to another in 3D world space. By minimizing the displacement error between skeletons in 3D space, we can exploit the epipolar geometry as well as the world-space and image-space position of joints to retrieve both the extrinsic parameters for rotation and translation $R | \mathbf{t}$ and the fundamental matrices $F_{\{i \rightarrow 0 | i=1, \dots, n\}}$.

A big advantage of the model we propose is the possibility to re-calibrate the network when the distances between the detected 3D skeletons diverges. Another feature is the capability of continuously improving calibration over time, for example in case new cameras are added, thanks due to the proposed 3D matching approach.

In the following paragraphs we denote the camera matrix P , while the intrinsic and extrinsic parameters are modelled by K and $[R | \mathbf{t}]$ respectively.

$$P = \left(\begin{array}{ccc|c} \overbrace{f_x}^K & s & x_0 & \\ 0 & f_y & y_0 & \\ 0 & 0 & 1 & \end{array} \right) \times \left(\begin{array}{cc|c} \overbrace{R}^{[R | \mathbf{t}]} & \overbrace{\mathbf{t}} & \\ \hline 0 & 1 & \end{array} \right) \quad (1)$$



Figure 3: 2D pose estimation for bounding box extraction

3.1 2D pose matching

The first module of our architecture takes as input n sequences of RGB frames and produces as output n bounding boxes of the pedestrian with the overall highest detection confidence score among all the views and with matching skeletal configuration. To ensure a robust detection, we employ the method described in [2] for joint parts and pairs detection. In a single pedestrian scenario and by relaxing the occlusions or noise constraints, even the classic vision background subtraction methods or some refined versions such as [17] can be used.

Although the skeletal information can be already exploited at this point to estimate the fundamental matrices, our proposal computes the matrices after re-projecting the 3D-world skeleton onto the image planes for more accurate results. To do so, we keep a reference to the displacement of the central point $[D_x^{pix}, D_y^{pix}]$ and the pixel-size of each bounding box, as well as an unscale factor. Note that in order to obtain the most accurate results possible, our framework accepts as input the prior knowledge of the people heights. However, approximate results can be obtained by providing an estimation of the known average height.

3.2 Mesh recovery

Each scaled bounding box $B_0, \dots, B_n$ contains now a crop of the original frames, containing the same pose of the subject, as seen from different viewpoints. Our goal is to jointly retrieve the 3D skeleton position from each viewpoint.

To achieve this, we employ a modified version of the method described in [9]. By feeding each bounding box B_i to the network, we obtain the vector Θ , corresponding to the SMPL [10] human body model parameters and configured as follows:

$$\Theta = \left[\overbrace{s, t_x, t_y}^{\text{camera}}, \overbrace{\theta}^{\text{pose}}, \overbrace{\beta}^{\text{shape}} \right] \quad (2)$$

From each human mesh $M(\theta, \beta)_i$ it is possible to obtain a set of $J = 19$ world-scale 3D joints ξ_i . The 10 body shape parameters β are used to refine the weak 2D pose matching described in 3.1 and remove remaining outliers. We discard the camera parameters s, t_x, t_y since they model a weak perspective pinhole camera model

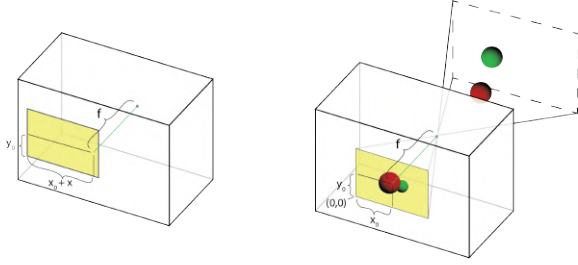


Figure 4: Principal point offset for mesh positioning in the weak perspective camera model does not correspond to a real-world mesh translation [16].

with its principal point shifted by $[t_x, t_y]$ (Fig. 4). In this model, the world translation of the mesh is computed as $z = F/s$, where s is a scaling factor.

The weak perspective model is not accurate enough for retrieving the real-world mesh transformation because it does not take into account perspective transformations. For this reason, we recover the real-world mesh displacement Δ^{mm} in millimeters, based on the perspective camera model:

$$\Delta^{mm} = \begin{bmatrix} \overbrace{\Delta_z^{mm} \Delta_x^{pix}}^{f_x^{pix}}, & \overbrace{\Delta_z^{mm} \Delta_y^{pix}}^{f_y^{pix}}, & \overbrace{f^{mm} w B^{mm}}^{W B^{pix}} \end{bmatrix} \quad (3)$$

where $f^{pix} = [f_x^{pix}, f_y^{pix}]$ corresponds to the focal length values in pixels, w is the image width in pixels and W is the sensor width in millimeters. B^{pix} and B^{mm} are the image-coordinates and world-coordinates sizes of the bounding boxes retrieved from 3.1. Now Δ_i^{mm} contains the real-world relative translation from C_i to ξ_i .

3.3 Skeleton matching

As of now, we built n camera-centric systems with n different 3D skeletons. We need to find the rotation matrices that map each skeleton $\xi_1, \dots, \xi_n$ to ξ_0 . We achieve this by considering a skeleton-centric system, in which each skeleton centroid c is placed in the world center. In this space, we can find the relative skeleton-to-skeleton transformations in terms of rotation and translation by a single value decomposition (SVD) of H , as explained in equations 4 and 5.

$$H = \xi_0 \cdot \xi_i \quad , \quad U, S, V^T = SVD(H) \quad (4)$$

$$R = V \cdot U^T \quad , \quad \mathbf{t} = c_i - (R \cdot c_0^T) \quad (5)$$

By starting from the skeleton-centric space, we apply the transformations of eq. 5 to ξ_i . Then, we move back to the camera-centric space and find the transformation that maps ξ_0 to Δ_0^{mm} . We finally find the inverse transformations Δ_i^{mm} , starting from eq. 3.

By applying this procedure, we obtain a 3D space in which C_0 is placed in the center of the coordinate system, the n skeletons are in Δ_0^{mm} and the relative position of all the other virtual cameras is

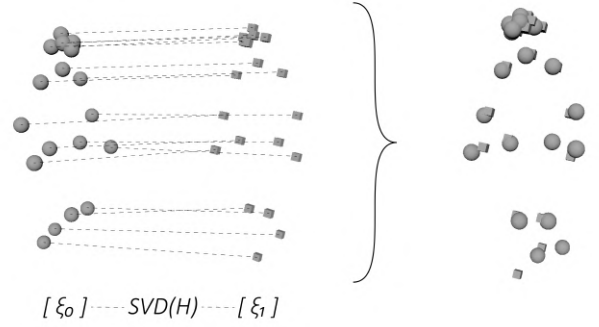


Figure 5: 3D skeleton matching for global transformation

known. A simple representation of the 3D skeleton match can be seen in Figure 5.

3.4 Fundamental matrix

With the skeletons $\xi_0, \dots, \xi_i$ correctly positioned in the 3D world, we calculate Σ as the 3D skeleton containing the mean values of all the joints in world-space coordinates. Since we also know the position of each camera in the 3D world, we can project Σ to each image plane of C_i , obtaining σ_i . We then build a vector σ_i containing 2D skeleton joints values for a batch of frames coming from C_i and use it as ordered keypoints to find the fundamental matrices $F_{i \rightarrow 0}$ that match camera C_i with camera C_0 .

This allows us to find the epipolar lines and corresponding matching points between two camera views. Moreover, since the extrinsic matrices have been previously retrieved, it is possible to describe how points in world coordinates map to each camera coordinate system, and viceversa.

4 RESULTS

We conducted three real-world experiments to test our framework in three different scenarios with different configurations, as listed in Table 1. The first experiment is similar to the setup provided in [4], even if it has been carried out at longer distances and with regular IP cameras instead of RGBD sensors. The second experiment considers the possibility of using videos from two arbitrary smartphones, to show how our method can also simultaneously work with two totally different sensors. The third scenario has been carried out inside a small indoor space, with two IP cameras facing one each other and where the views suffer partial occlusions due to the environment configuration. It is worth noting that our method also works with a single camera, since the depth is being estimated directly from the 3D skeleton extracted from each individual frame.

4.1 Quantitative results

The main results of our experiments are listed in Table 1 and Table 2. As can be seen, they are comparable with the results provided by [4], particularly taking into consideration that we only employ monocular cameras and no additional depth sensors. The metrics MinSDE, ASDE and MaxSDE, describe the minimum, average, and maximum displacement of skeletal joints, respectively, after the matching in

3D space, in meters, calculated by the Euclidean distance:

$$SDE = \sqrt{\sum_{i=1}^n (\xi_0 - \xi_i)^2} \quad (6)$$

RPD, VPD (real and virtual plane displacements) are the measures of the displacement from the origin along the real world plane and the virtual world plane respectively. The RPD has been calculated starting from ground truth annotations, while the VPD can be calculated once again with an Euclidean distance from the origin, discarding the z component. The plane displacement error (PDE) is computed as $|RPD - VPD|$, once again in meters. The MRE is the mean reprojection error calculated after applying the fundamental matrix F to the set of points σ_i .

Our results for scenario 1 and 2 are also better than the checkerboard results obtained by [4] using the method described in [23].

	Experiments						
	Gym			Laboratory		Apartment	
Configuration							
Cameras	3			2		2	
Frames	250			250		875	
Sensor size	1/3"			1/3.2"	1/2.8"	1/2.8"	
	9.1 mm			5.7 mm	6.3 mm	7 mm	
Focal length	3.0	3.0	5.5	4.0	4.07	4.5	6
3D Matching							
MinSDE	0.04			0.01		0.10	
ASDE	0.08			0.06		0.13	
MaxSDE	0.14			0.08		0.16	
Real-world displacement							
RPD	7.73			5.66		3.26	
VPD	7.92			5.79		3.35	
PDE	0.21			0.13		0.09	

Table 1: Experimental results. Columns: the three different test scenarios. Rows: number of cameras, number of frames, 3D skeleton displacement error (min, average, max), displacement along the plane (real plane, virtual plane, plane displacement error)

4.2 Reprojection error

After finding the fundamental matrices F for each scene and the corresponding epipolar lines, we assess the precision of our method by calculating the reprojection error in term of point-line-distance, as follows:

$$\frac{|ax_0 + bx_0 + c|}{\sqrt{a^2 + b^2}} \quad (7)$$

where a , b and c are the epipolar lines coefficients and $[x_0, y_0]$ are the coordinates of the projected points. In table 2 the reprojection errors in pixels for each FullHD scene are listed, showing that the proposed method is robust in all the three test scenarios considered.

	Experiments		
	Gym	Laboratory	Apartment
MRE (pix)	6.15	12.03	8.38

Table 2: Reprojection errors for the three test environments

4.3 Qualitative results

In Figures 6, 7, 8 we provide some qualitative results through the Autodesk Maya® 3D animation, modeling, simulation, and rendering software. In each image a reconstruction of the 3D scene is shown, including the 3D skeleton used for the matching, the virtual plane and every virtual camera with correct roll, pitch, yaw, translation, focal length and frustum size. We decided to discard the approximate differentiable render OpenDR [11] used by [10] and [9] in favour of Maya because the latter lets us configure in fine details many camera parameters such as the focal length, the film gate and frustum size in millimeters. Moreover, our entire code can directly run into the Maya environment, allowing us to easily extend the scope of our work to weak monocular 3D human motion capture from video footage, also from a single camera. Our 3D reconstruction module in Maya is standalone and can receive camera and skeleton data from an external machine via a command port socket.

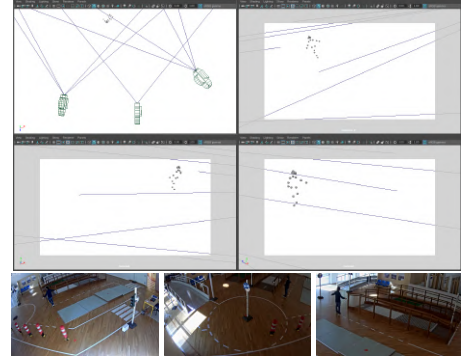


Figure 6: Gym scenario

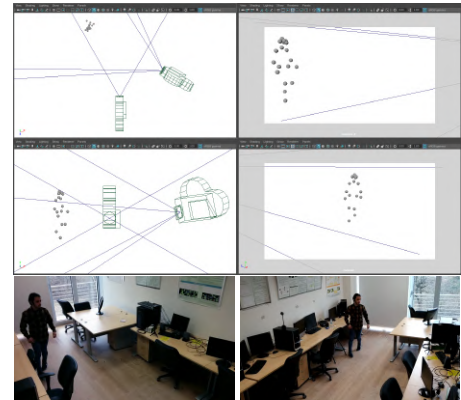


Figure 7: Laboratory scenario

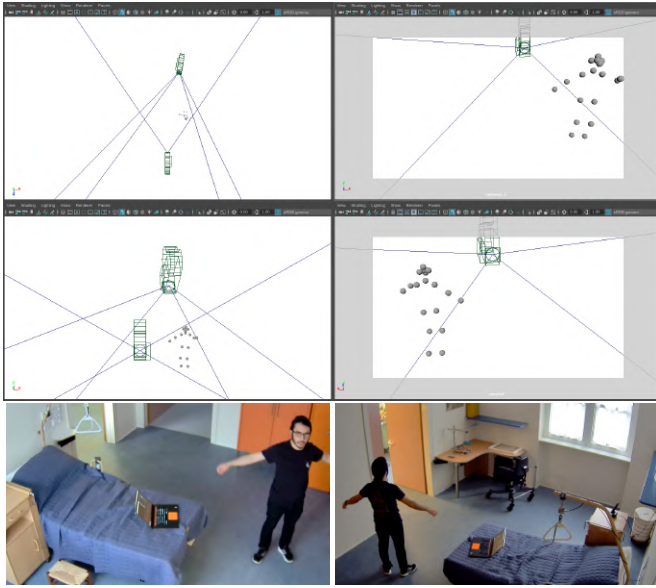


Figure 8: Apartment scenario

5 CONCLUSIONS

We presented our framework for a fully unsupervised markerless multi-camera calibration through monocular 3D human pose estimation. It consists of a 3-stage approach which employs (i) 2D human detection and matching between views, (ii) monocular 3D pose estimation through mesh recovery and (iii) joint 2D and 3D skeleton matching in camera-centric and skeleton-centric spaces. The overall output of our work consists of the extrinsic parameters that link cameras coordinates with world coordinates, as well as all the fundamental matrices to link the camera views. Compared to the other related works in literature, our approach works with at least one monocular RGB sensor and it is hardware-independent. It is robust to occlusions and noise in the scene thanks to the skeleton matching approach, and it is able to re-calibrate the camera network in real-time. Additionally, its accuracy can be refined over time in a continuous fashion.

5.1 Future work

As future work we plan to explore some solutions to improve the framework accuracy, such as introducing different skeletal configurations or discarding some sets of joints during the 3D matching. We plan to adopt some mesh matching algorithms from the computer graphics field to improve the accuracy of the skeleton matching described in section 3.3. We also plan to extend our work to intrinsic parameters estimation in order to provide a complete pipeline for calibration and distortion correction.

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To my dear friends who always support me.

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